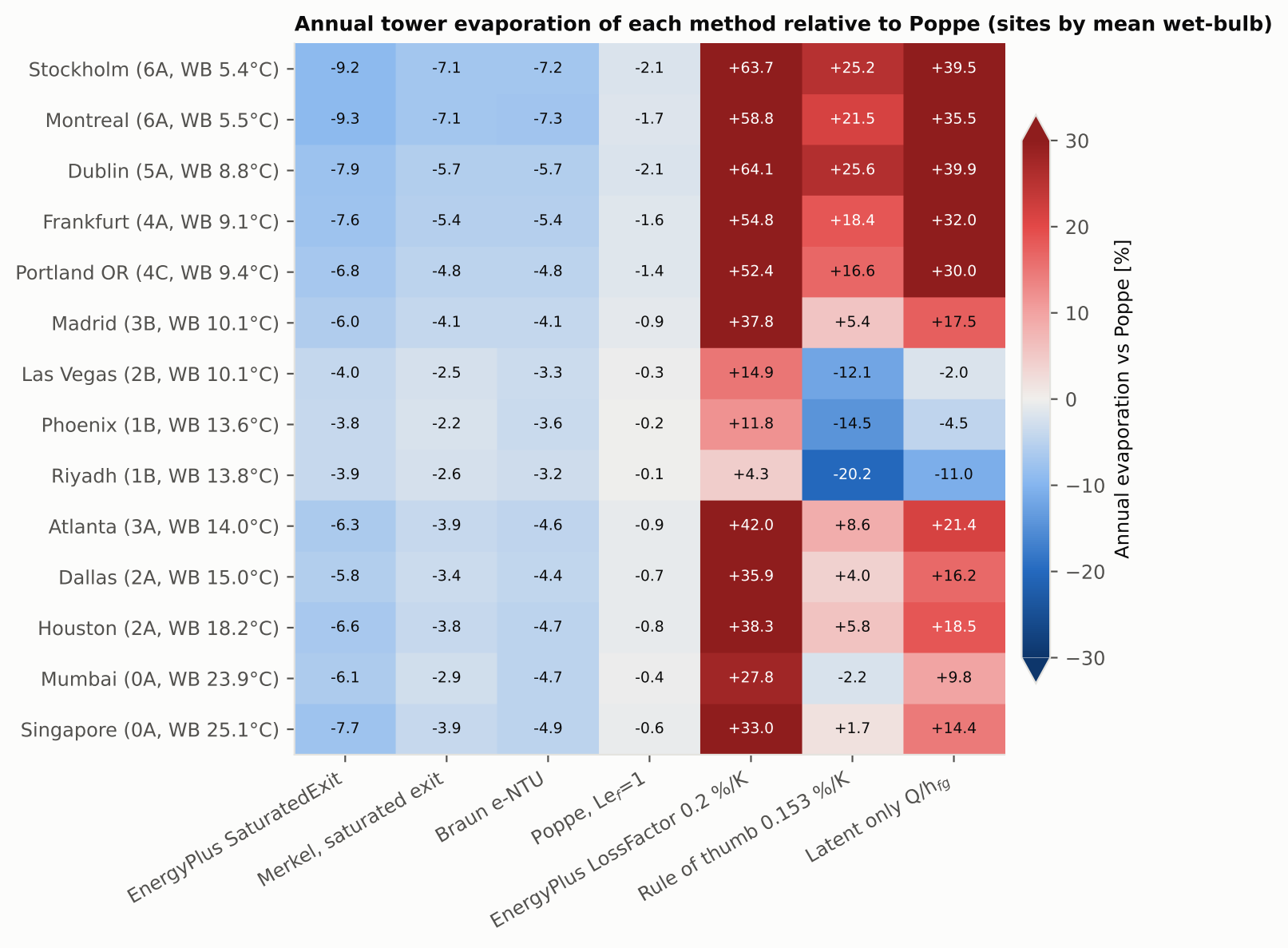

Annual tower evaporation of each method relative to Poppe (sites by mean wet-bulb)
Stockholm (6A, WB 5.4°C)
Montreal (6A, WB 5.5°C)
Dublin (5A, WB 8.8°C)
Frankfurt (4A, WB 9.1°C)
Portland OR (4C, WB 9.4°C)
Madrid (3B, WB 10.1°C)
Las Vegas (2B, WB 10.1°C)
Phoenix (1B, WB 13.6°C)
Riyadh (1B, WB 13.8°C)
Atlanta (3A, WB 14.0°C)
Dallas (2A, WB 15.0°C)
Houston (2A, WB 18.2°C)
Mumbai (0A, WB 23.9°C)
Singapore (0A, WB 25.1°C)
-9.2 -7.1 -7.2 -2.1 +63.7 +25.2 +39.5
-9.3 -7.1 -7.3 -1.7 +58.8 +21.5 +35.5
-7.9 -5.7 -5.7 -2.1 +64.1 +25.6 +39.9
-7.6 -5.4 -5.4 -1.6 +54.8 +18.4 +32.0
-6.8 -4.8 -4.8 -1.4 +52.4 +16.6 +30.0
-6.0 -4.1 -4.1 -0.9 +37.8 +5.4 +17.5
-4.0 -2.5 -3.3 -0.3 +14.9 -12.1 -2.0
-3.8 -2.2 -3.6 -0.2 +11.8 -14.5 -4.5
-3.9 -2.6 -3.2 -0.1 +4.3 -20.2 -11.0
-6.3 -3.9 -4.6 -0.9 +42.0 +8.6 +21.4
-5.8 -3.4 -4.4 -0.7 +35.9 +4.0 +16.2
-6.6 -3.8 -4.7 -0.8 +38.3 +5.8 +18.5
-6.1 -2.9 -4.7 -0.4 +27.8 -2.2 +9.8
-7.7 -3.9 -4.9 -0.6 +33.0 +1.7 +14.4
EnergyPlus SaturatedExit
Merkel, saturated exit
Braun e-NTU
Poppe, Le$_f$=1
EnergyPlus LossFactor 0.2 %/K
Rule of thumb 0.153 %/K
Latent only Q/h$_{fg}$
Annual evaporation vs Poppe [%]
30
20
10
0
−10
−20
−30